

Pedro Mariano

Finance & Process Automation | Python · SQL · AI

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Open to hybrid roles in Lisbon or remote roles

SUMMARY

Business Management student at ISCAL (accounting, auditing, Portuguese corporate and personal tax) who builds automation on real finance data. At my family's minimarket I built a pricing system that reads supplier invoices, proposes costs and retail prices, and writes approved changes into the store's back-office database with backups, transactions and an audit log. I also built a DeFi lending protocol in Solidity (97.8% test coverage) and an AI red-teaming tool whose findings map to the EU AI Act. Available now for a full-time hybrid role in business or finance systems analysis, finance automation, ERP or tax technology.

EXPERIENCE

Operations & Systems — family minimarket, Loures

Jan 2026 – Present

- Run day-to-day operations and the store's systems: back-office in MS Access, point-of-sale (POS) checkout on SQL Server, 16,840 products and sales history since 2007.
- Built a pricing system in Python and PowerShell (about 3,300 lines) that integrates the back-office with the POS database. It reads supplier invoices from photos using OCR, the Portuguese fiscal QR code and line-total checks, then proposes costs and prices.
- Made every database write safe: a verified backup first, a single transaction, an automatic abort if the record changed after it was read, and a log of previous values.
- Tested every pricing rule on real invoices with the owner before go-live (UAT), wrote test cases for edge cases, and turned each defect found by two independent code audits into a regression test.
- Built a read-only margin report that ranks below-cost products by money lost over 90 days, using audit materiality (ISA 320), without slowing the checkout.
- Prepared the service contract, GDPR processing terms (art. 28) and NDA to offer the system to other stores that use the same back-office software (about 600 stores).

Founder (side venture) — AI red-teaming for customer-facing chatbots, Lisbon

2026

- Built an automated tester in Python and Azure OpenAI that sends adversarial prompts to AI assistants and scores findings across about 16 risk categories, such as PII leakage, prompt injection and system-prompt leaks.
- Mapped every finding to EU AI Act articles so a compliance team could use the report as audit evidence, and defined the pricing and sales outreach for mid-sized EU companies.
- Closed the venture after assessing entry barriers and better-funded competitors, and shifted focus to finance automation.

PROJECTS

Invoice → Price demo (public)

github.com/Limin-design/invoice-pricing-demo

- Public rewrite of the store pricing system on synthetic data: fiscal QR proof per VAT rate, pricing rules, safe writes with undo, 22 tests, a Power BI dashboard and an n8n workflow that triages new invoices.

CryptoPawn — lending protocol for tokenized stocks (Solidity, Hardhat, JavaScript)

2026

- Designed USDC loans against tokenized US stocks, gold and BTC, with interest schedules, late fees and collateral rules written as a specification for external auditors.
- Wrote 75+ automated tests (97.8% line coverage) and randomized invariant tests, including runs against the real AAPLx token on an Ethereum mainnet fork.
- Found and fixed a critical accounting bug: the real token rebases for dividends and splits, which the first version did not handle. Deployed to a public testnet.

StratForge — trading strategy backtester (TypeScript, React, Tauri)

github.com/Limin-design/stratforge · 2026

- Built a desktop app that backtests trading strategies with fees and slippage, and checks whether results are robust or overfit (Monte Carlo, walk-forward, out-of-sample, deflated Sharpe ratio).
- Added an AI analyst that calls the deterministic engine instead of computing results itself, with API keys encrypted on the device.

SKILLS

- **Finance & tax:** Portuguese accounting standards (SNC/NCRF), personal and corporate income tax (IRS, IRC), SAF-T (PT), financial auditing (ISA 320 materiality, internal control), margin and cost analysis, financial modeling (P&L, scenarios, unit economics)
- **Automation & data:** Python (pandas), SQL (SQL Server, MS Access/Jet), REST API and systems integration, PowerShell, Excel, Power BI (data model, DAX), OCR and document data extraction, ETL and data validation, n8n workflows, scheduled jobs
- **Testing & QA:** test cases, regression testing, user acceptance testing (UAT), bug investigation, pytest
- **Business analysis:** requirements gathering, process mapping, functional specifications, ERP and finance processes (procure-to-pay, supplier invoices, pricing and margin control), user documentation
- **Software:** JavaScript/TypeScript, React, Node.js, Solidity (Hardhat, OpenZeppelin), Go, Redis (NoSQL), Git, Docker
- **AI:** Microsoft Azure (resource setup, Azure OpenAI deployments, Azure AI Search indexes, API key management), LLM APIs (Claude, Gemini), RAG and semantic search with Azure AI Search, AI agents and agentic workflows (Claude Code, n8n), rapid prototyping and proofs of concept, local model training and fine-tuning, including data preparation for efficient training, prompt engineering, AI red-teaming and evaluation, EU AI Act
- **Languages:** Portuguese (native), Romanian (native), English (fluent, C1)

EDUCATION

Bachelor's in Business Management (Licenciatura em Gestão) — ISCAL, Lisbon 2023 – expected 2027
Average grade: 15/20 (Portuguese 0–20 scale). Relevant coursework: financial accounting (SNC), financial auditing, Portuguese taxation (IRS, IRC), human resources management, entrepreneurship.

Secondary education, Economics — Escola Secundária José Afonso, Loures 2018 – 2021